Mukesh Kanna R

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Full-Stack Developer | React.js, Next.js, Node.js, MongoDB, MySQL, TypeScript (MERN) | AWS | AI-Assisted Development

PROFESSIONAL SUMMARY

Results-driven Full Stack Developer with 3+ years of experience building scalable, high-performance web applications using the MERN stack (MongoDB, Express.js, React.js, Node.js) and Next.js. Proven expertise in architecting RESTful APIs, implementing Role-Based Access Control (RBAC) with JWT authentication, and delivering multi-tenant SaaS platforms across healthcare and edtech domains. Skilled in AI-assisted development (vibe coding) with Claude Code, Cursor, GitHub Copilot and ChatGPT to accelerate delivery, debugging and documentation while keeping code quality and security under developer review.

TECHNICAL SKILLS

- **Frontend:** React.js, Next.js (SSR/SSG/Server Actions), Vue.js, TypeScript, JavaScript (ES6+), HTML5, CSS3, Tailwind CSS, Redux, Context API, React Hooks
- **Backend:** Node.js, Express.js, Python, REST API Design, Middleware, MVC Architecture, Microservices Architecture, Server Actions
- **Database:** MongoDB, Mongoose, MySQL, Aggregation Pipelines, Database Schema Design
- **Cloud:** AWS (EC2, S3)
- **Auth & Security:** JWT Authentication, Role-Based Access Control (RBAC), Multi-tenant Architecture
- **Healthcare Integration:** HL7, Malaffi (Abu Dhabi HIE), EMR Regulatory Compliance
- **AI Tools:** Claude Code, ChatGPT, GitHub Copilot, Cursor, Agentic Coding Workflow
- **AI-Assisted Development (Vibe Coding):** Prompt engineering for code generation, AI pair programming, agentic coding workflows, AI-driven debugging and refactoring, AI-assisted code review, test and documentation generation, rapid prototyping
- **Tools & CMS:** Strapi CMS, Git, GitHub, Postman, JIRA, Trello, VS Code
- **Concepts:** API Testing, Dynamic Routing, Responsive UI Design, Agile/Scrum

WORK EXPERIENCE

Full Stack Engineer – Neurula Technologies **May 2026 – Present**

- Working on the EMR (Electronic Medical Record) platform, contributing to core modules and UAE/Abu Dhabi regulatory compliance features.
- Building and maintaining full-stack features using React.js, Next.js, Node.js and related MERN-stack technologies.
- Using Claude Code for agentic development and refactoring, reducing turnaround on feature and integration work.
- Collaborating with cross-functional teams to deliver scalable healthcare platform capabilities.

Full-Stack Developer – TechParrot Innovations LLP **Dec 2025 – May 2026**

- Architected a multi-tenant EdTech platform (Vidhyaan) with RBAC, JWT auth and isolated data access across 10+ institutions.
- Built fee management, student enrollment and analytics dashboards using React.js, Next.js, Node.js, Express.js and MySQL.
- Designed RESTful APIs and scalable multi-tenant architecture with secure per-institution data handling.

Application Developer – Aspigrow Technologies **Jul 2023 – Nov 2025**

- Built 3 production web apps (IntegrityCE, Med Marshal, Health Marshal) across healthcare and edtech using React.js, Next.js and Node.js.
- Implemented JWT authentication, RBAC and RESTful APIs; integrated Strapi CMS for content management.
- Designed reusable UI components with Tailwind CSS and built admin dashboards for real-time data monitoring.

KEY PROJECTS

1) IntegrityCE – Medical College Examination & Academic Portal

Medical college exam portal with SSR admin panel, 35+ REST APIs and Strapi CMS integration.

My Contributions as Developer:

- Built the SSR admin panel in Next.js + TypeScript for exam scheduling, results publishing and academic content.
- Developed 35+ REST APIs in Node.js/Express.js backed by MySQL for exams, students and results.
- Integrated Strapi CMS so non-technical staff can manage academic content without code deployments.
- Implemented JWT authentication with RBAC for admin, faculty and student roles; built reusable Tailwind CSS components.

AI Tools Used: GitHub Copilot, ChatGPT

How I Used AI: Scaffolded Express routes and TypeScript interfaces, generated Strapi content-type configs, and debugged Next.js SSR hydration issues with AI pair programming.

Tech: React.js, Next.js (SSR), TypeScript, Node.js, Express.js, MySQL, Strapi CMS, Tailwind CSS

2) Med Marshal – Medical Laboratory Management Platform

Multi-lab healthcare platform with centralized doctor/patient management, appointments, revenue and inventory tracking.

My Contributions as Developer:

- Designed Mongoose schemas for labs, doctors, patients, appointments and inventory.
- Built appointment scheduling and patient records modules end-to-end (UI, APIs, database).
- Wrote MongoDB aggregation pipelines powering revenue and inventory analytics dashboards.
- Developed REST APIs enabling centralized management across multiple labs.

AI Tools Used: GitHub Copilot, ChatGPT, Cursor

How I Used AI: Used AI to draft and optimize complex aggregation pipelines, generate API test cases, and rapidly prototype dashboard UI components.

Tech: React.js, Next.js, Node.js, Express.js, MongoDB, Mongoose, REST APIs

3) Vidhyaan – EdTech Multi-Institution Management Platform

Multi-tenant EdTech platform with student admission, fee management, RBAC and institutional analytics.

My Contributions as Developer:

- Architected the multi-tenant design with isolated per-institution data access across 80+ institutions.
- Implemented JWT authentication and RBAC for super admin, institution admin, staff and student roles.
- Built admission, fee management and institutional analytics dashboard modules.
- Designed the MySQL schema and RESTful APIs with secure per-tenant query scoping.

AI Tools Used: Claude, Cursor, GitHub Copilot

How I Used AI: Used vibe coding workflows to prototype modules quickly, reviewed multi-tenant query logic for data-leak risks with AI, and generated SQL migrations and API documentation.

Tech: React.js, Next.js, Node.js, Express.js, MySQL, JWT, RBAC

4) Neurula Healthcare EMR – Electronic Medical Record Platform

EMR platform serving healthcare facilities in the UAE, built to Abu Dhabi regulatory standards.

My Contributions as Developer:

- Building role-based dashboards (Nurse, Doctor, Admin, etc...) tailored to clinical and administrative workflows.
- Designing and documenting the Finance & Accounting module for the EMR.
- Implementing UAE/Abu Dhabi regulatory compliance features, including HL7/Malaffi integration for school health screening forms.
- Writing Python scripts for data processing and HL7 message integration.

AI Tools Used: Claude Code, Claude, GitHub Copilot

How I Used AI: Use Claude Code in VS Code for agentic feature development, codebase exploration, refactoring and HL7 message mapping; use AI to draft technical and compliance documentation.

Tech: React.js, Next.js, Node.js, Express.js, MySQL, Python, HL7, JWT, RBAC, Tailwind CSS

CERTIFICATIONS

- Getting Started with DevOps on AWS — Amazon Web Services
- Complete Git and GitHub Course — Udemy

EDUCATION

B.E – Computer Science and Engineering Aug 2019 – May 2023
Muthayammal Engineering College

HSC Jun 2018 – Apr 2019
Vedhha Vikass Boys Higher Secondary School

SSLC Jun 2016 – Apr 2017
S.R.V Boys Higher Secondary School